

A → Differential (Because in option)

B → Shunt (Because constant flux)

C → Cumulative

09/07/14

* SPEED CONTROL →

$$N \propto \frac{E_b}{\phi} \propto \frac{V - I_a R_a}{\phi} \rightarrow \text{vary the speed}$$

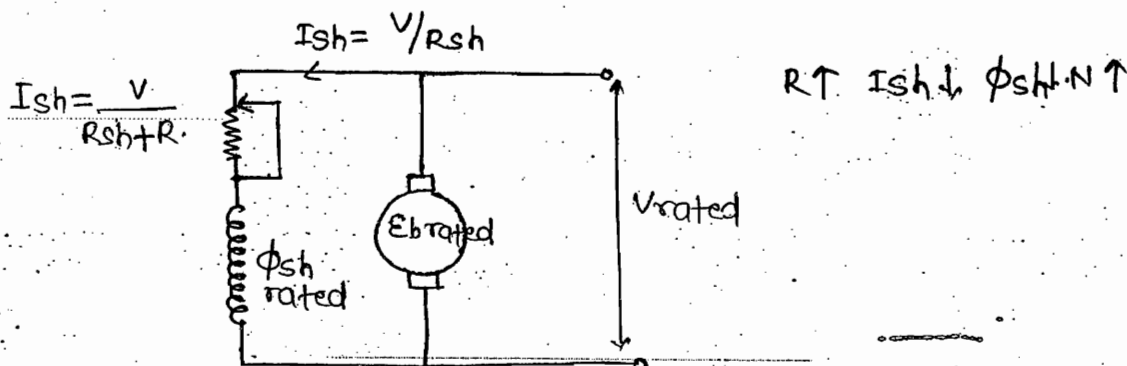
(1) Flux - Field weakening

(2) R_a - Arm. Resistance / Rheostat

(3) V - Voltage Control.

* Speed control of shunt motor →

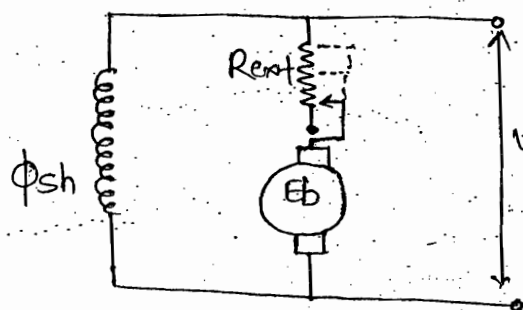
(1) Field weakening / Field current →



- * By connecting an external resistance in series with field wdg & by increasing it from its min^m to max^m the field current will reduce & reduce its flux & increase its speed.
- * The min^m speed in this method is only rated.
- * It is efficient speed control method.
- * Because here we decrease the value of flux (field weakening)
- * Loss associated with external resistance in the field ckt is small
- * No need of large rheostat
- * No need of additional cooling.

- * Requires 4-Point starter specially to control speed over wide range.
- * Additional arm. reaction as the main flux is reduced. Therefore it requires interpole, compensating wdg etc.
- * Also called as variable torque constant power speed control.
- * Field resistance should be in the min^m resistance position during starting.
- * There is a limit to decrease the field current or to increase the field resistance as the speed of motor becomes dangerously high.

(2) Rheostatic / Arm. Resistance Control →



$$\downarrow E_b = [V - I_a(R_a + R_{ext})] \downarrow$$

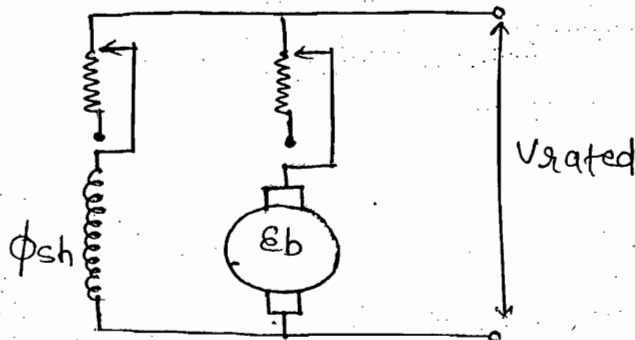
$$\downarrow \frac{E_b}{\Phi} \propto N \downarrow$$

- * By connecting an external resistance in series with arm. the voltage across the motor arm. is varied consequently the speed is vary.
- * It is basically the vol. control without varying supply vol.
- * It includes resistance & produce more cu loss. Therefore less efficient method.
- * It is to control speeds from rated to below.
- * No need of 4 point starter & there is no additional arm. reaction effect.
- * Requires additional cooling methods.

* Also called constant torque variable power.

* The arm. Rheostat should be at its maxm position during starting.

Both above Speed control →



FWSC

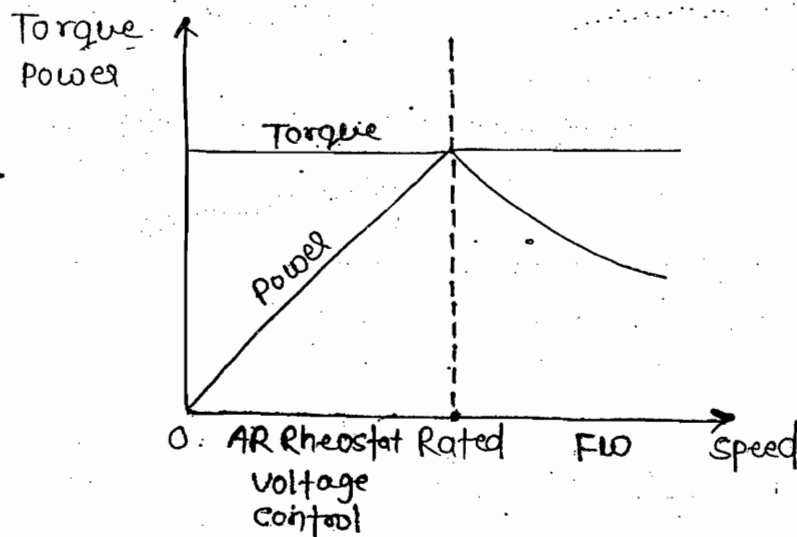
* Variable torque
Constant power

* N_{rated} to above

ARSC

Constant torque
Variable power

Rated to below



Ques → A shunt motor is running at rated speed 1000 rpm with Rated voltage V . If the voltage becomes half ($V/2$) then the speed will

- ① 1000 ② 500 ③ 250 ④ 2000

Soln →

$$\frac{N_2}{N_1} = \frac{E_{b2}}{E_{b1}} \times \frac{\phi_1}{\phi_2}$$

$$\frac{N_2}{N_1} = \frac{E_{b1/2}}{E_{b1}} \times \frac{\phi_1}{\phi_{1/2}}$$

$$\frac{N_2}{N_1} = 1$$

$$N_2 = 1000 \text{ RPM}$$

$$\left\{ \begin{aligned} N &\propto \frac{E_b}{\phi} \propto \frac{V - I_a R_a}{\phi} \\ E_b &\propto \frac{V}{\phi} \quad (I_a R_a \text{ small}) \end{aligned} \right.$$

$$I_{sh} = \frac{V_{sh}}{R_{sh}}$$

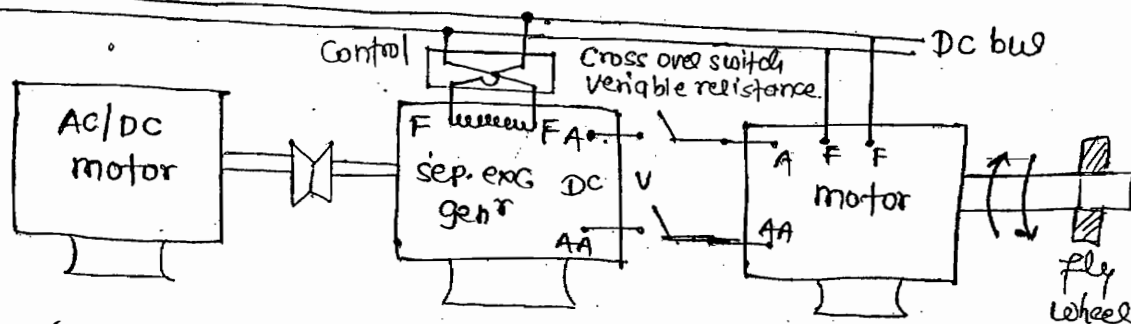
$$\left\{ \begin{aligned} \text{If } V = V/2, I_{sh} &= I_{sh}/2 \\ \text{then } \phi &= \phi_{sh}/2 \end{aligned} \right.$$

Case ② If the flux is constant

$$N_2 = 500 \text{ RPM}$$

* Multiple voltage control →

- * self excited shunt motor should be separately excited in order to apply the speed control tech.
- * It requires a multiple vol. dc source.
- * For large rating motor specially used under steel rolling voltage control method known as Ward-Leonard speed control is used which requires separately excited dc gen^r & the motor to rotate the gen^r.

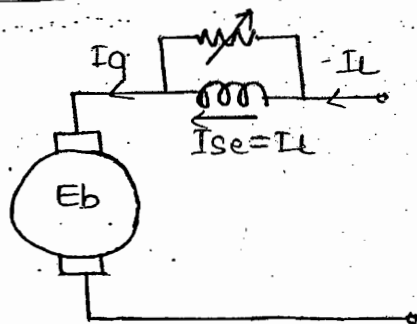


- * This is exclusively use to control the speed from rated to below rated efficiently without inserting the resistance in series with arm. & to control speed in both dirn.
- * The control is through a cross over switch which varies the excitation of separately excited gen^r.
- * As it requires two additional m/c it is very expensive.
- * A flywheel is connected across the shaft.
- * In order to control the speed fluctuations due to varying loads & to improve overall ~~flg~~ efficiency known as Ward Leonard Ilgner method.

* Speed control of series motors →

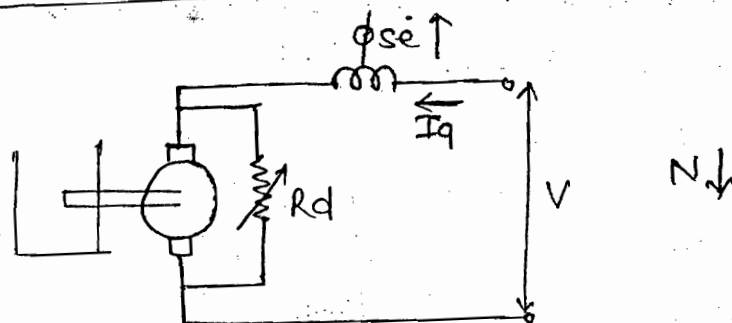
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|-----------------------------|-------------------------|
| (1) Field diverter | (5) Armature Resistance |
| (2) Armature diverter | (6) Multiple voltage. |
| (3) Tapped field | |
| (4) Paralleling Field coils | |

(1) Field diverter →



- * By connecting a diverter the current flowing through the field wdg gets diverted, flux reduce & consequently speed ↑.
- * Diverter resistance should not reduce below a min^m value as the speed becomes dangerously high.

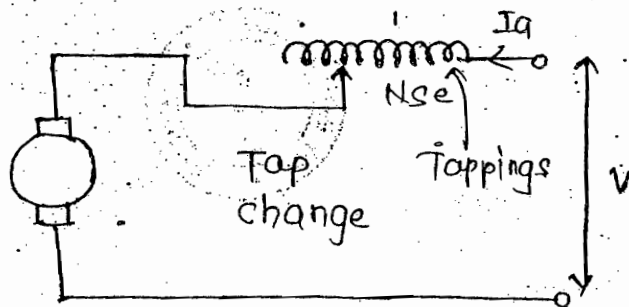
② Armature diverter →



$$T \propto \phi I_a$$

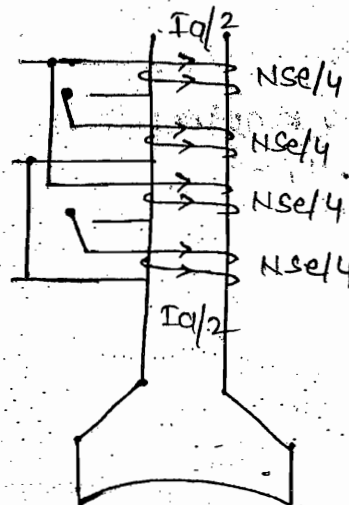
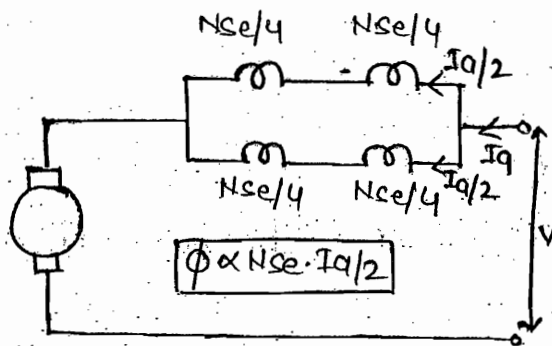
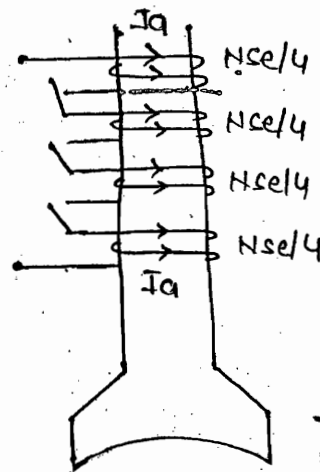
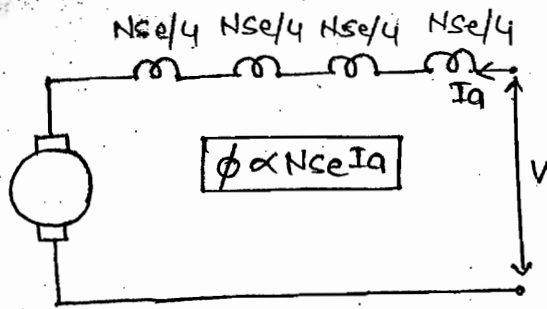
- * By connecting diverter across arm., arm. current gets diverted due to which the torque is reduced,
- * In order to maintain the torque the motor draws more current from supply which comes through the field wdg
Flux $\uparrow$ Speed $\downarrow$.

③ Tapped Field →



- * If the tappings are more, then turns are more & hence by increasing the value of turns the value of flux increases & the speed may decrease.

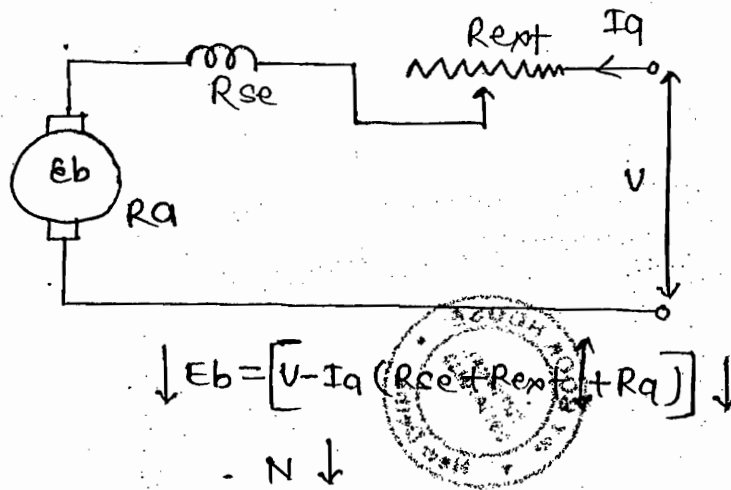
(4) Paralleling Field Coils →



* The field wdg is arranged in the form of coils which can be externally switch in series & parallel.

* In order to get speeds b/w the steps a diverter may be connected

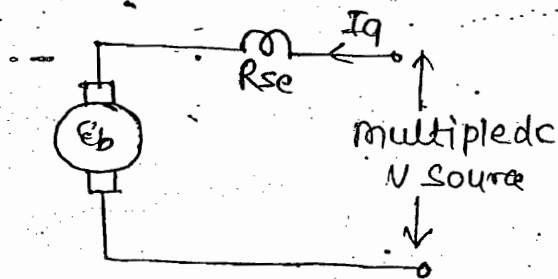
⑤ Armature Resistance Control →



* By adjusting the resistance, vol. across the arm. is vary to vary the speed. Which involves huge power loss.

* It also requires a multiple vol. source. Therefore these are rarely used.

⑥ Multiple vol. Control →



* Traction motors in series & parallel →

